Vikash Vashisth

CFD Specialist | Multiphase and Reactive Flows | Thermal-Fluid and Multiphysics Simulation

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PROFESSIONAL SUMMARY

CFD specialist with a PhD in Chemical Engineering, 4+ years of post-PhD experience and over 9 years of CFD and multiphysics experience. Advanced ANSYS Fluent practitioner across multiphase flow, LES/RANS turbulence, reactive flows, species transport, heat and mass transfer, rotating flows and transient simulation. Experienced across the complete CFD workflow, from problem definition, meshing and numerical verification to experimental validation, HPC execution and engineering recommendations. Leads CFD model development in an industry-academic collaboration and communicates complex results to technical and industrial stakeholders.

CORE EXPERTISE

CFD physics: Multiphase flow and VOF; reactive flow; finite-rate chemistry; species transport; heat and mass transfer; thermal radiation; plasma reactors; non-Newtonian and free-surface flow; rotating equipment.

Turbulence and numerics: RANS; LES; transient simulation; sliding mesh; adaptive mesh refinement; mesh and time-step studies; convergence assessment; numerical verification; experimental validation; HPC and parallel computing.

Software and automation: ANSYS Fluent; ANSYS Workbench; ANSYS Meshing and DesignModeler; Siemens NX - working knowledge; ANSYS HFSS - working knowledge; STAR-CCM+ - working knowledge; PyFluent; Python; MATLAB; UDF/C; Tecplot; Linux.

Engineering delivery: Problem scoping; simulation methodology development; validation planning; design and operating recommendations; calculation documentation; project-team and stakeholder presentations; collaboration with industrial and experimental partners; technical project leadership; technical reporting; MSc and intern supervision.

PROFESSIONAL EXPERIENCE

Chalmers University of Technology — **Gothenburg, Sweden**

Researcher - CFD and Multiphysics — **Dec 2025 - Present**

Postdoctoral Researcher - CFD and Multiphysics — **Dec 2022 - Dec 2025**

- **Microwave plasma reactor:** Lead 3D CFD model development for a 2.45 GHz reactor, coupling flow, heat transfer, species transport and finite-rate chemistry to evaluate temperature fields and reactor performance across operating conditions.
- **Verification and validation:** Developed mesh-independent reactor and forced-quenching models; validated predicted gas and wall temperatures and outlet NOx concentrations against experimental measurements across operating conditions.
- **Quantified reactor performance:** Predicted a 6-10% increase in NO yield and a 6-8% reduction in specific energy consumption relative to corresponding non-quenched reference configurations, supporting design and operating-condition comparisons.
- **Multiphase method development:** Developed and validated transient LES-VOF models of turbulence-driven bubble and droplet breakup with adaptive mesh refinement; correlated breakup behaviour with high-speed experiments and achieved about 94% resolved turbulent kinetic energy.
- **Large-scale numerical execution:** Planned and executed an HPC campaign of about 2 million core-hours, quantifying interactions between turbulent structures and evolving fluid interfaces.
- **Resolved breakup physics:** Captured micrometer-scale interface evolution beyond the imaging resolution of the experiments and quantified breakup mechanisms, satellite formation and daughter-size distributions.
- **Technical leadership and presentation:** Lead biweekly CFD reviews with NitroCapt AB and research partners, translate results into engineering recommendations.
- **EM-CFD coupling:** Implemented a manual one-way HFSS-Fluent data-transfer workflow; automated two-way coupling and PyFluent integration are in development.

PROFESSIONAL EXPERIENCE - CONTINUED

IIT Guwahati - Reckitt Benckiser Collaboration **Guwahati, India**

Research Associate - Fixed-Term Industrial CFD Project **Jan 2022 - Apr 2022**

- **Industrial multiphase simulation:** Developed transient 3D VOF multiphase models of shear-thinning rivulet flow on inclined surfaces, coupling measured rheology with interface transport.
- **Industrial delivery:** Established mesh independence, validated wetted-area and propagation trends against experiments, and delivered technical findings to the industrial collaboration.

Indian Institute of Technology Roorkee **Roorkee, India**

Doctoral Researcher - Fluid Mechanics, Mixing and Rotating Flows **Dec 2015 - Apr 2021**

- **Rotor-stator CFD:** Developed validated transient sliding-mesh RANS and LES models for high-shear mixers, quantifying flow, pumping, power demand, energy dissipation, residence time distribution (RTD) and dead zones across conventional and tilted-hole stators.
- **Validation and design trade-offs:** Achieved power-number agreement within 10% of benchmark data and evaluated the balance between mixing performance and power consumption across geometries.
- **Numerical execution:** Built hybrid meshes up to 4.5 million cells and modelled tracer and species transport for Newtonian and non-Newtonian systems.
- **Mentoring:** Guided MSc CFD work on meshing, solver setup, post-processing and validation, contributing to peer-reviewed research outputs.

EDUCATION

PhD, Chemical Engineering - Indian Institute of Technology Roorkee **2015 - 2021**

Thesis: Hydrodynamics and Mixing Studies in High Shear Mixers for Viscous Fluids

Master of Technology (M.Tech), Chemical Engineering - Indian Institute of Technology Roorkee **2011 - 2013**

SELECTED PUBLICATIONS AND PROFESSIONAL EVIDENCE

Research output: Peer-reviewed publications across turbulence, multiphase flow, interface breakup, rotating equipment, reactor systems, numerical methods and experiment-simulation validation.

- Analysis of forced and intrinsic quenching in a plasma reactor for nitrogen fixation by use of three-dimensional simulations and experiments. Chemical Engineering Journal, 2026.
- Dynamics of bubble breakup under turbulent flow conditions. Chemical Engineering Journal, 2025.
- Analysis of a plasma reactor performance for direct nitrogen fixation by use of three-dimensional simulations and experiments. Chemical Engineering Journal, 2024.

Profiles: Google Scholar | ORCID

Professional service: Peer reviewer for Chemical Engineering Journal, Chemical Engineering Science, Chemical Engineering and Processing, Industrial & Engineering Chemistry Research and Results in Engineering.

Training and recognition: Ansys Learning Hub training in PyFluent and HFSS (2026); Residential Fellow, Institute of Advanced Studies, Loughborough University (2023).

Additional: English - fluent; Swedish - basic and actively learning; Swedish driving licence (B); open to relocation within Sweden.